Name: Sanjeev Kumar Patel **Enrollment No.:** 2024BCSE022

Lab 8

Q1: Using HTTPS, retrieve users (id, name, username, email) from a URL, store the data in an SQLite database, and display all the stored users.

```python
import requests
import sqlite3

url = "https://jsonplaceholder.typicode.com/users"

# HTTPS GET request
users = requests.get(url).json()

# SQLite database
con = sqlite3.connect("users.db")
cur = con.cursor()

# Create table
cur.execute("""
CREATE TABLE IF NOT EXISTS users (
    id INTEGER PRIMARY KEY,
    name TEXT,
    username TEXT,
    email TEXT
)
""")

# Store data
for user in users:
    cur.execute(
        "INSERT OR REPLACE INTO users VALUES (?, ?, ?, ?)",
        (user["id"], user["name"], user["username"], user["email"])
    )

con.commit()

# Display data
cur.execute("SELECT * FROM users")

print("ID | Name | Username | Email")
print("-" * 60)

for row in cur.fetchall():
    print(row)

```

```
41 con.close()
```

Output:

```
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab8> python .\
    httpsSql.py
ID | Name | Username | Email
--------------------------------------------------------------
(1, 'Leanne Graham', 'Bret', 'Sincere@april.biz')
(2, 'Ervin Howell', 'Antonette', 'Shanna@melissa.tv')
(3, 'Clementine Bauch', 'Samantha', 'Nathan@yesenia.net')
(4, 'Patricia Lebsack', 'Karianne', 'Julianne.OConner@kory.
    org')
(5, 'Chelsey Dietrich', 'Kamren', 'Lucio_Hettinger@annie.ca'
    )
(6, 'Mrs. Dennis Schulist', 'Leopoldo_Corkery', '
    Karley_Dach@jasper.info')
(7, 'Kurtis Weissnat', 'Elwyn.Skiles', 'Telly.Hoeger@billy.
    biz')
(8, 'Nicholas Runolfsdottir V', 'Maxime_Nienow', '
    Sherwood@rosamond.me')
(9, 'Glenna Reichert', 'Delphine', 'Chaim_McDermott@dana.io'
    )
(10, 'Clementina DuBuque', 'Moriah.Stanton', 'Rey.
    Padberg@karina.biz')
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab8>
```

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Lab 8

Q2: Using the JSONPlaceholder API, perform GET, POST, DELETE, and PUT operations. Retrieve a user's name and email using their username, create a new user, delete a user by ID, and update a user's information using their username.

```python
import requests

url = "https://jsonplaceholder.typicode.com/users"

# Using GET Method Search user by username
username = input("Username: ")
users = requests.get(url).json()

for u in users:
    if u["username"].lower() == username.lower():
        print(u["name"], u["email"])

# POST - Create new user
user = {
    "name": input("Name: "),
    "username": input("Username: "),
    "email": input("Email: ")
}

response = requests.post(url, json=user)
print("Created:", response.json())

# DELETE - Delete user
user_id = input("User ID to delete: ")

response = requests.delete(f"{url}/{user_id}")
print("Delete status:", response.status_code)

# PUT - Update user
username = input("Username to update: ")

for u in users:
    if u["username"].lower() == username.lower():

        updated = {
            "name": input("New name: "),
            "username": u["username"],
            "email": input("New email: ")
```

```
        }

        response = requests.put(
            f"{url}/{u['id']}",
            json=updated
        )

        print("Updated:", response.json())
        break
```

Output:

```
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab8> python .\
    jsonPLaceholder.py
Username: Leanne
Name: Bret
Username: Leanne Graham
Email: sa@gamil
Created: {'name': 'Bret', 'username': 'Leanne Graham', '
    email': 'sa@gamil', 'id': 11}
User ID to delete: 2
Delete status: 200
Username to update: Leanne Graham
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab8>
```

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Lab 8

Q3: Create a student database using Python and MySQL. Insert five students using executemany(). Display all students, display CSE students, display students having marks greater than 85, update marks of a student, and delete a student.

```
import sqlite3

# Connect to database
db = sqlite3.connect("student.db")
cur = db.cursor()

# Create table
cur.execute("""
CREATE TABLE IF NOT EXISTS students (
    id INTEGER PRIMARY KEY,
    name TEXT,
    branch TEXT,
    marks INTEGER
)
""")

# Insert 5 students using executemany()
students = [
    (1, "Rahul", "CSE", 90),
    (2, "Priya", "ECE", 82),
    (3, "Aman", "CSE", 88),
    (4, "Neha", "ME", 75),
    (5, "Riya", "CSE", 92)
]

cur.executemany(
    "INSERT INTO students VALUES (?, ?, ?, ?)",
    students
)
db.commit()

# Display all students
print("\nAll Students:")
cur.execute("SELECT * FROM students")
for row in cur.fetchall():
    print(row)

# Display CSE students
```

```
print("\nCSE Students:")
cur.execute("SELECT * FROM students WHERE branch = ?", ("CSE",))
for row in cur.fetchall():
    print(row)

# Display students with marks > 85
print("\nStudents with marks > 85:")
cur.execute("SELECT * FROM students WHERE marks > ?", (85,))
for row in cur.fetchall():
    print(row)

# Update marks
student_id = int(input("\nEnter student ID: "))
new_marks = int(input("Enter new marks: "))

cur.execute(
    "UPDATE students SET marks = ? WHERE id = ?",
    (new_marks, student_id)
)
db.commit()

print("Marks updated successfully.")

# Delete a student
student_id = int(input("\nEnter student ID to delete: "))

cur.execute(
    "DELETE FROM students WHERE id = ?",
    (student_id,)
)
db.commit()

print("Student deleted successfully.")

db.close()

db = sqlite3.connect("student.db")
cur = db.cursor()

print("\nCSE Students:")
cur.execute("SELECT * FROM students WHERE branch = ?", ("CSE",))
for row in cur.fetchall():
    print(row)
db.close()
```

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Output:

```
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab8> python .\
    studentDatabase.py

All Students:
(1, 'Rahul', 'CSE', 90)
(2, 'Priya', 'ECE', 82)
(3, 'Aman', 'CSE', 88)
(4, 'Neha', 'ME', 75)
(5, 'Riya', 'CSE', 92)

CSE Students:
(1, 'Rahul', 'CSE', 90)
(3, 'Aman', 'CSE', 88)
(5, 'Riya', 'CSE', 92)

Students with marks > 85:
(1, 'Rahul', 'CSE', 90)
(3, 'Aman', 'CSE', 88)
(5, 'Riya', 'CSE', 92)

Enter student ID: 2
Enter new marks: 95
Marks updated successfully.

Enter student ID to delete: 4
Student deleted successfully.

CSE Students:
(1, 'Rahul', 'CSE', 90)
(3, 'Aman', 'CSE', 88)
(5, 'Riya', 'CSE', 92)
PS S:\SANJEEV PATEL\College\3rd year\Python\Lab8>
```